

The Trigonometry of Collapse

Angular Geometry, Phase Dynamics, and
Rotational Structure in Complex System Instability

A Complete Mathematical Treatment of the
Canonical-BSDT Angular Framework

Most mathematical frameworks for complex system failure monitor scalar magnitudes: energy, distance, variance, likelihood. This book develops a complementary view: collapse is fundamentally an angular phenomenon. Systems do not simply grow unstable — they lose their direction. The gradient and the force cease to point the same way. Agents synchronise their phases into a collective that the restoring geometry cannot hold. The critical manifold is not a threshold of magnitude but a surface where curvature overwhelms gradient. This book derives these statements rigorously, building from the three fundamental angles of the canonical framework to a complete trigonometric architecture for early warning, phase prediction, and collapse characterisation.

Contents

I	The Three Fundamental Angles	1
1	The Alignment Angle θ_t	2
1.1	Definition and geometric origin	2
1.2	The energy derivative in terms of θ_t	2
1.3	Force decomposition via θ_t	3
1.4	Angular velocity and acceleration	3
1.5	The alignment angle as a stability indicator	3
1.6	Relationship between θ_t and the admissibility threshold	4
2	The Misalignment Angle ψ_t	5
2.1	Definition	5
2.2	The Jacobian as an angle-generating map	5
2.3	Why ψ_t resolves false alarms	6
2.4	The two MFLS quantities and the amplification ratio	6
3	The Collapse Angle $\tan \theta_t$	7
3.1	Definition	7
3.2	Relation to $\cos \theta_t$	7
3.3	Three equivalent collapse conditions	7
3.4	The $\tan \theta - \lambda_{\max}$ bridge	8
3.5	Relationship among all three angles	8
II	Rotational Geometry	9
4	$SO(3)$ and the Lie Group Canonical Engine	10
4.1	Rotations as canonical states	10
4.2	The Rodrigues formula: trigonometry of rotation	10

4.3	The canonical gradient in $\text{SO}(3)$	11
4.4	Rotation angle as energy proxy	11
4.5	The canonical ODE on $\text{SO}(3)$	11
5	Symplectic Geometry as Angle-Preserving Structure	12
5.1	The symplectic form and orthogonality	12
5.2	Hamiltonian dynamics as angle-preserving flow	12
5.3	The canonical projection preserves Hamiltonian	12
5.4	Phase space as a torus near equilibrium	13
III	Phase Dynamics	14
6	The Phase-Amplitude Decomposition	15
6.1	Decomposing state space into amplitude and direction	15
6.2	Amplitude dynamics	15
6.3	Phase dynamics	16
7	The Kuramoto Order Parameter as Canonical Channel	17
7.1	Phase extraction	17
7.2	The Kuramoto order parameter	17
7.3	Phase coherence channel δ_Φ	17
7.4	Phase energy as canonical Lyapunov function	18
7.5	Phase gain and phase-aware ODE	18
IV	The Three-Phase Precursor	19
8	The Precursor Theorem and Its Proof	20
8.1	Statement	20
8.2	Proof	20
8.3	Temporal ordering of the phases	21
8.4	Falsifiability	21
V	Trigonometric Curvature	22
9	The Hessian as a Curvature Tensor	23
9.1	Second-order geometry of the energy landscape	23
9.2	The spectral radius as a curvature indicator	23
9.3	Curvature flow: the Ricci connection	24
10	Geodesics and the Critical Manifold	25
10.1	Geodesics on energy level sets	25
10.2	The critical manifold as a curvature threshold surface	25
10.3	Angle of approach to the critical manifold	25
VI	Harmonic Structure and Spectral Collapse	27
11	Fourier Decomposition of Canonical Trajectories	28

11.1	Harmonic representation	28
11.2	Instability as spectral shift	28
11.3	Resonance and cascade	28
12	Graph Wavelets as Angular Multi-Scale Decomposition	30
12.1	The wavelet band as an angle	30
12.2	Hausdorff dimension as an angular invariant	30
VII	Domain Applications	31
13	Power Grid: Phase-Angle Synchronisation Collapse	32
13.1	The ERCOT phase portrait as trigonometric collapse	32
13.2	The ERCOT +1033h lead in trigonometric terms	32
13.3	The Casimir as a phase constraint	33
14	EEG: Seizure as Phase Coherence Transition	34
14.1	The alignment angle as a seizure precursor	34
14.2	Angular signatures in the alignment-commitment manifold	34
15	Crypto Futures: Angular Trading Signals	35
15.1	The v58 modulator as an angular filter	35
15.2	The $\ddot{\theta}_t > 0$ entry signal	35
15.3	The price loading factor as an angular projection	35
VIII	Complete Angular Reference	37
16	All Angles, All Formulas	38
16.1	The seven canonical angles	38
16.2	Trigonometric identities in canonical form	38
16.3	The three-phase precursor in angular notation	39
16.4	Honest scope summary	40
IX	The Ellipsoidal Origin	41
17	The Ellipsoid Was Already the Energy	42
17.1	The original ellipsoidal geometry	42
17.2	The canonical energy is the same object	42
17.3	The surface-normal angle is the alignment angle	43
17.4	Two states with the same ρ but different futures	43
17.5	The adaptive γ origins	43
17.6	The complete developmental chain	44
17.7	The dual description theorem	44
17.8	Summary: what was always there	45

Part I

The Three Fundamental Angles

Chapter 1

The Alignment Angle θ_t

1.1 Definition and geometric origin

The canonical dynamical-geometry engine operates by projecting the nominal force field F_{base} onto the energy gradient $g_X = 2J^\top GS$. The quality of this alignment determines the engine's effectiveness. The alignment angle quantifies it.

Definition 1.1 (Alignment angle).

$$\cos \theta_t = \frac{\langle g_X(t), F_{\text{base}}(t) \rangle}{\|g_X(t)\| \|F_{\text{base}}(t)\|} \in [-1, +1], \quad \theta_t = \arccos(\cos \theta_t) \in [0, \pi].$$

$\theta_t = 0$: F_{base} points exactly along g_X — nominal force fully drives instability.
 $\theta_t = \pi/2$: $F_{\text{base}} \perp g_X$ — force tangential, no first-order energy change.
 $\theta_t = \pi$: F_{base} anti-parallel to g_X — force actively opposes instability.

1.2 The energy derivative in terms of θ_t

Theorem 1.2 (Energy derivative via alignment).

$$\dot{E} = (1 - \gamma) \|F_{\text{base}}\| \|g_X\| \cos \theta_t - (1 - \gamma) \|g_X\|^2.$$

Proof. From the canonical ODE, $\dot{E} = (1 - \gamma)[\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2]$. Substituting $\langle g_X, F_{\text{base}} \rangle = \|g_X\| \|F_{\text{base}}\| \cos \theta_t$ gives the result. $\square$

Corollary 1.3 (Descent condition in angular form).

$$\dot{E} \leq 0 \iff \cos \theta_t \leq \frac{\|g_X\|}{\|F_{\text{base}}\|} = \frac{\text{MFLS}}{\|F_{\text{base}}\|}.$$

Descent is guaranteed whenever the alignment angle exceeds the critical value $\theta_t^ = \arccos(\text{MFLS}/\|F_{\text{base}}\|)$.*

1.3 Force decomposition via θ_t

Proposition 1.4 (Force decomposition). *The nominal force decomposes into collapse-directed and safe components:*

$$F_{\text{base}} = F_{\parallel} + F_{\perp}, \quad F_{\parallel} = \|F_{\text{base}}\| \cos \theta_t \hat{u}, \quad F_{\perp} = F_{\text{base}} - \|F_{\text{base}}\| \cos \theta_t \hat{u},$$

where $\hat{u} = g_X / \|g_X\|$. Then:

$$\|F_{\parallel}\| = \|F_{\text{base}}\| \cos \theta_t, \quad \|F_{\perp}\| = \|F_{\text{base}}\| \sin \theta_t, \quad \|F_{\text{base}}\|^2 = \|F_{\parallel}\|^2 + \|F_{\perp}\|^2.$$

The canonical ODE damps $F_{\parallel}$ by factor $(1 - \gamma)$ and leaves $F_{\perp}$ free.

Proof. $\|F_{\parallel}\|^2 = \|F_{\text{base}}\|^2 \cos^2 \theta_t$, $\|F_{\perp}\|^2 = \|F_{\text{base}}\|^2 - \|F_{\parallel}\|^2 = \|F_{\text{base}}\|^2 \sin^2 \theta_t$. Pythagorean identity: $\cos^2 \theta_t + \sin^2 \theta_t = 1$. $\square$

1.4 Angular velocity and acceleration

Theorem 1.5 (Angular velocity, exact formula). *Let $h(t) = \cos \theta_t$, $p = \langle g_X, F_{\text{base}} \rangle$, $a = \|g_X\|$, $b = \|F_{\text{base}}\|$. Then $\dot{\theta}_t = -\dot{h} / \sin \theta_t$ where:*

$$\dot{h} = \frac{\langle \dot{g}_X, F_{\text{base}} \rangle + \langle g_X, \dot{F}_{\text{base}} \rangle}{ab} - \cos \theta_t \left(\frac{\langle g_X, \dot{g}_X \rangle}{a^2} + \frac{\langle F_{\text{base}}, \dot{F}_{\text{base}} \rangle}{b^2} \right).$$

Proof. Differentiate $h = p/(ab)$ by the quotient rule: $\dot{h} = (\dot{p} \cdot ab - p(\dot{a}b + a\dot{b})) / (ab)^2 = \dot{p}/(ab) - h(\dot{a}/a + \dot{b}/b)$. Then $\dot{p} = \langle \dot{g}_X, F_{\text{base}} \rangle + \langle g_X, \dot{F}_{\text{base}} \rangle$, $\dot{a} = \langle g_X, \dot{g}_X \rangle / a$, $\dot{b} = \langle F_{\text{base}}, \dot{F}_{\text{base}} \rangle / b$. Substitute to obtain the stated formula. Since $\theta_t = \arccos(h)$, $\dot{\theta}_t = -\dot{h} / \sqrt{1 - h^2} = -\dot{h} / \sin \theta_t$ for $\theta_t \in (0, \pi)$. $\square$

Definition 1.6 (Angular acceleration).

$$\ddot{\theta}_t = \frac{d^2 \theta_t}{dt^2} = \frac{d \dot{\theta}_t}{dt}.$$

In discrete time: $\ddot{\theta}_t \approx (\theta_{t+1} - 2\theta_t + \theta_{t-1}) / \Delta t^2$.

Proposition 1.7 (Sign conventions). $\dot{\theta}_t > 0$: angle opening, force rotating away from gradient.
 $\dot{\theta}_t < 0$: angle closing, force aligning with gradient.
 $\ddot{\theta}_t > 0$: decoherence accelerating.
 $\ddot{\theta}_t < 0$: decoherence decelerating (angular dynamics stabilising).

1.5 The alignment angle as a stability indicator

Theorem 1.8 (Safety ratio and alignment). *The safety ratio satisfies:*

$$\text{SafetyRatio}(t) = \frac{\theta \text{MFLS}_t}{2M_{\max} \sqrt{E_t}}.$$

Its time derivative is:

$$\frac{d}{dt}\text{SafetyRatio} = \frac{\theta}{2M_{\max}} \frac{M\dot{F}LS_t\sqrt{E_t} - MFLS_t\dot{E}_t/(2\sqrt{E_t})}{E_t}.$$

The safety ratio falls whenever $M\dot{F}LS_t/MFLS_t < \dot{E}_t/(2E_t)$, i.e. gradient grows slower than energy.

Proof. Differentiate $\text{SafetyRatio} = \theta MFLS/(2M_{\max}\sqrt{E})$ by the quotient rule. The condition $d\text{SafetyRatio}/dt < 0$ rearranges to $2E_t M\dot{F}LS_t < MFLS_t \dot{E}_t$, i.e. $M\dot{F}LS_t/MFLS_t < \dot{E}_t/(2E_t)$. $\square$

1.6 Relationship between θ_t and the admissibility threshold

From $\Psi^* = \theta MFLS/(4\sqrt{E})$ and the descent condition $\cos \theta_t \leq MFLS/\|F_{\text{base}}\|$:

Corollary 1.9 (Angular admissibility). *The system is admissible ($M_{\max} < \Psi^*$) if and only if:*

$$\cos \theta_t^* \leq \frac{2M_{\max}\sqrt{E}}{\theta \|F_{\text{base}}\|}$$

at the critical alignment angle. The system breaches admissibility when the alignment angle falls below this threshold simultaneously with energy rising.

Chapter 2

The Misalignment Angle ψ_t

2.1 Definition

The canonical ODE uses the state-space gradient $\tilde{G}_t = J^\top g_t$ (the pullback of the channel-space gradient through the Jacobian). The original BSDT master operator used the channel-space gradient g_t directly. These two vectors are generally not parallel. The angle between them is ψ_t .

Definition 2.1 (Misalignment angle).

$$\cos \psi_t = \frac{\langle \tilde{G}_t, g_t \rangle_F}{\|\tilde{G}_t\|_F \|g_t\|} = \frac{R_t}{\|\tilde{G}_t\|_F \|g_t\|}$$

where $\tilde{G}_t = J_t^\top g_t \in \mathbb{R}^{Nd}$ is the pullback gradient, $g_t = \nabla_S E_{BS}(S_t) \in \mathbb{R}^4$ is the channel-space gradient, and $R_t = g_t^\top \mathcal{G}_t g_t = g_t^\top J_t J_t^\top g_t \geq 0$ is the control coupling sum.

2.2 The Jacobian as an angle-generating map

The Jacobian $J_t \in \mathbb{R}^{4 \times Nd}$ maps channel-space vectors to state-space vectors: $J_t^\top : \mathbb{R}^4 \rightarrow \mathbb{R}^{Nd}$.

Proposition 2.2 (Singular value decomposition of J). *Let $J_t = U \Sigma V^\top$ be the thin SVD with $U \in \mathbb{R}^{4 \times 4}$, $\Sigma \in \mathbb{R}^{4 \times 4}$, $V \in \mathbb{R}^{Nd \times 4}$. Then:*

$$\tilde{G}_t = J_t^\top g_t = V \Sigma U^\top g_t.$$

The misalignment angle satisfies:

$$\cos \psi_t = \frac{(U^\top g_t)^\top \Sigma^2 (U^\top g_t)}{\|V \Sigma U^\top g_t\| \|g_t\|} = \frac{\|\Sigma U^\top g_t\|^2}{\|\Sigma U^\top g_t\| \|g_t\|} = \frac{\|\Sigma U^\top g_t\|}{\|g_t\|}.$$

Proof. $R_t = g_t^\top J J^\top g_t = \|J^\top g_t\|^2 = \|V \Sigma U^\top g_t\|^2 = \|\Sigma U^\top g_t\|^2$ (since V is orthonormal).
 $\|\tilde{G}_t\|_F = \|J^\top g_t\| = \|\Sigma U^\top g_t\|$. Substituting: $\cos \psi_t = R_t / (\|\tilde{G}_t\| \|g_t\|) = \|\Sigma U^\top g_t\|^2 / (\|\Sigma U^\top g_t\| \|g_t\|) = \|\Sigma U^\top g_t\| / \|g_t\|$. $\square$

Corollary 2.3 (Singular value formula for ψ_t).

$$\cos \psi_t = \frac{\sqrt{\sum_{k=1}^4 \sigma_k^2(J_t) [U^\top g_t]_k^2}}{\|g_t\|}.$$

The misalignment angle depends on how the channel-space gradient g_t aligns with the singular vectors of J_t and on the singular values σ_k .

2.3 Why ψ_t resolves false alarms

Theorem 2.4 (ψ_t as false-alarm discriminator). Let $\|g_t\|$ (channel-space MFLS) be large. Then:

- (i) If $\psi_t \approx 0$: $\cos \psi_t \approx 1$, so $\|\tilde{G}_t\| \approx \|g_t\|$. Both channel-space and state-space intensities are large. The alarm is genuine: the physical dynamics are aligned with the risk signal.
- (ii) If $\psi_t \approx \pi/2$: $\cos \psi_t \approx 0$, so $\|\tilde{G}_t\| \approx 0$. Channel-space intensity is large but state-space intensity is small. The alarm is a false positive: abstract risk elevated, physical dynamics tangential.

Proof. $\|\tilde{G}_t\| = \|J^\top g_t\| = \|g_t\| \cos \psi_t$ (from Definition 2.1). If $\cos \psi_t \approx 0$ then $\|\tilde{G}_t\| \approx 0$ regardless of $\|g_t\|$. The canonical ODE correction is proportional to $\|\tilde{G}_t\|^2$; with $\|\tilde{G}_t\| \approx 0$ the correction vanishes and the physical system is unaffected. $\square$

2.4 The two MFLS quantities and the amplification ratio

Definition 2.5 (Dual MFLS).

$$\text{MFLS}^{\text{channel}}(t) = \|g_t\| \quad (\text{abstract intensity}), \quad \text{MFLS}^{\text{state}}(t) = \|\tilde{G}_t\|_F \quad (\text{physical intensity}).$$

$$\rho_{\text{MFLS}}(t) = \frac{\text{MFLS}^{\text{state}}}{\text{MFLS}^{\text{channel}}} = \cos \psi_t \in [0, 1].$$

$\rho_{\text{MFLS}} > 0.7$: physical system amplifies the abstract risk signal.

$\rho_{\text{MFLS}} < 0.4$: risk is geometrically contained despite high channel score.

Proof. $\text{MFLS}^{\text{state}}/\text{MFLS}^{\text{channel}} = \|J^\top g_t\| / \|g_t\| = \cos \psi_t$ from the singular value formula. $\square$

Chapter 3

The Collapse Angle $\tan \theta_t$

3.1 Definition

Definition 3.1 (Collapse angle).

$$\tan \theta_t = \frac{\|F_{\perp}\|}{\|F_{\parallel}\|} = \frac{\|F_{\text{base}}\| \sin \theta_t}{\|F_{\text{base}}\| \cos \theta_t} = \frac{\sin \theta_t}{\cos \theta_t}.$$

$\tan \theta_t \rightarrow 0$: force fully collapse-directed, all energy is driving instability.

$\tan \theta_t \rightarrow \infty$: force fully perpendicular, no first-order energy change.

$\tan \theta_t = 1$ ($\theta_t = 45^\circ$): critical balance.

3.2 Relation to $\cos \theta_t$

Proposition 3.2 (Fundamental trigonometric identities).

$$\sin \theta_t = \sqrt{1 - \cos^2 \theta_t}, \quad \tan \theta_t = \frac{\sqrt{1 - \cos^2 \theta_t}}{\cos \theta_t}, \quad \sec \theta_t = \frac{1}{\cos \theta_t}.$$

These are exact; no approximation is involved.

3.3 Three equivalent collapse conditions

Theorem 3.3 (Triple equivalence of collapse conditions). *The following three conditions are equivalent at the critical manifold $\mathcal{C}_{\text{man}}$:*

- (I) **Spectral:** $\lambda_{\max}(\nabla^2 \Phi(X_t)) > 1$
- (II) **Angular:** $\tan \theta_t < \tan \theta_t^*$ where $\tan \theta_t^* = \sqrt{\|F_{\text{base}}\|^2 - \gamma^{*2} \text{MFLS}^2} / (\gamma^* \text{MFLS})$
- (III) **Energetic:** $\lambda_{\max}(\nabla^2 E_{\text{BS}}) > \text{MFLS}^2 / E$

Proof. (I) $\iff$ (III): The Hessian of the canonical energy $\nabla^2 E_{\text{BS}} = 2A + \text{diag}(\beta_k^2 e^{\beta_k S_k})$. Supercriticality $\lambda_{\max} > 1$ means the curvature exceeds the gradient norm per unit energy.

(I) $\iff$ (II): The angular condition parameterises the Hessian threshold through the force decomposition: at the critical manifold the parallel force $\gamma^* \text{MFLS}$ equals the spectral threshold times the perpendicular force. Substituting gives the stated $\tan \theta^*$. $\square$

3.4 The $\tan \theta$ – $\lambda_{\max}$ bridge

Theorem 3.4 ($\tan \theta$ connects gradient, curvature, and spectral threshold). *Let v_1 be the principal eigenvector of $\nabla^2 E_{\text{BS}}$ and φ the angle between ∇E_{BS} and v_1 . Then:*

$$\tan \theta_{\text{curv}} = \frac{\text{MFLS} \sin \varphi}{\lambda_{\max}(\nabla^2 E_{\text{BS}}) \cos \varphi}.$$

When $\varphi \rightarrow 0$ (gradient aligns with principal curvature): $\tan \theta_{\text{curv}} \rightarrow 0$ — collapse is curvature-dominated. When $\varphi \rightarrow \pi/2$: $\tan \theta_{\text{curv}} \rightarrow \infty$ — curvature irrelevant.

Proof. The second-order energy change along direction v is $v^\top \nabla^2 E_{\text{BS}} v$. Decompose $\nabla E_{\text{BS}} = c_1 v_1 + v_\perp$ where $c_1 = \text{MFLS} \cos \varphi$ and $\|v_\perp\| = \text{MFLS} \sin \varphi$. The curvature-to-gradient ratio along v_1 is $\lambda_{\max} c_1 / \text{MFLS} = \lambda_{\max} \cos \varphi$. The tangential component is $\text{MFLS} \sin \varphi$. Their ratio is the stated $\tan \theta_{\text{curv}}$. $\square$

3.5 Relationship among all three angles

The three angles satisfy a fundamental geometric identity:

Theorem 3.5 (Angular triangle). *At any state X_t :*

$$\theta_t + \psi_t + \varphi_t \leq \pi$$

where θ_t is the alignment angle, ψ_t is the misalignment angle, and φ_t is the curvature-gradient angle. Equality holds when all three vectors (F_{base} , $\tilde{G}_t$, v_1) are coplanar.

Proof. Each angle is defined between vectors in $\mathbb{R}^{N_d}$. For three vectors in a common plane, the three pairwise angles sum to π (the triangle inequality becomes an equality). In general (non-coplanar vectors), the sum is less. The result follows from the spherical triangle inequality on the unit sphere in $\mathbb{R}^{N_d}$. $\square$

Part II

Rotational Geometry

Chapter 4

$SO(3)$ and the Lie Group Canonical Engine

4.1 Rotations as canonical states

Let $X = R \in SO(3)$, a 3×3 rotation matrix. The Lie algebra is $\mathfrak{so}(3) = \{W \in \mathbb{R}^{3 \times 3} : W = -W^T\}$ (skew-symmetric matrices).

Every $W \in \mathfrak{so}(3)$ corresponds to a vector $\omega \in \mathbb{R}^3$ via:

$$W = \hat{\omega} = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}, \quad \|\hat{\omega}\|_F = \sqrt{2} \|\omega\|.$$

4.2 The Rodrigues formula: trigonometry of rotation

Theorem 4.1 (Rodrigues' rotation formula). *For $\omega \in \mathbb{R}^3$ with $\|\omega\| = \theta > 0$ and axis $\hat{n} = \omega/\theta$:*

$$\exp(\hat{\omega}) = I + \sin \theta \hat{n} + (1 - \cos \theta) \hat{n}^2.$$

Proof. The matrix exponential $\exp(\hat{\omega}) = \sum_{k=0}^{\infty} \hat{\omega}^k / k!$. For skew-symmetric $\hat{n}$ ($\|\hat{n}\| = 1$): $\hat{n}^3 = -\hat{n}$, $\hat{n}^4 = -\hat{n}^2$, $\hat{n}^5 = \hat{n}$, etc. Separating even and odd powers: $\exp(\theta \hat{n}) = I + (\theta - \theta^3/6 + \dots) \hat{n} + (\theta^2/2 - \theta^4/24 + \dots) \hat{n}^2 = I + \sin \theta \hat{n} + (1 - \cos \theta) \hat{n}^2$. $\square$

Corollary 4.2 (Trace formula). $\text{tr}(\exp(\hat{\omega})) = 1 + 2 \cos \theta$. *The rotation angle is recoverable from the trace: $\theta = \arccos((\text{tr}(R) - 1)/2)$.*

Proof. $\text{tr}(I) = 3$, $\text{tr}(\hat{n}) = 0$, $\text{tr}(\hat{n}^2) = -2$ (for unit-norm skew-symmetric $\hat{n}$). Thus $\text{tr}(\exp(\hat{\omega})) = 3 + 0 + (1 - \cos \theta)(-2) = 1 + 2 \cos \theta$. $\square$

4.3 The canonical gradient in $\text{SO}(3)$

For $R \in \text{SO}(3)$ and feature map $S : \text{SO}(3) \rightarrow \mathbb{R}^k$, the left-trivialised Jacobian is:

$$\hat{J}(R) = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} S(R \exp(\varepsilon \cdot)) : \mathfrak{so}(3) \rightarrow \mathbb{R}^k.$$

Proposition 4.3 (Lie group canonical gradient). *The canonical gradient in $\text{SO}(3)$ is:*

$$g_{\text{so}(3)} = 2\hat{J}^*GS \in \mathfrak{so}(3)$$

where $\hat{J}^*$ is the metric adjoint: $\langle \hat{J}^*v, \xi \rangle_{\mathfrak{so}(3)} = \langle v, \hat{J}\xi \rangle_{\mathbb{R}^k}$.

Proof. This is the $\text{SO}(3)$ instantiation of the abstract canonical gradient $g_X = 2J^\top GS$, with J replaced by the left-trivialised Jacobian $\hat{J}$ and the inner product on $\mathbb{R}^n$ replaced by the Lie algebra metric. The metric adjoint plays the role of $J^\top$. $\square$

4.4 Rotation angle as energy proxy

Theorem 4.4 (Energy as squared rotation angle). *For $S(R) = \log(R^\top R_{\text{target}})$ (the logarithm of the relative rotation), $G = I$:*

$$E(R) = \|S(R)\|_F^2 = \left\| \log(R^\top R_{\text{target}}) \right\|_F^2 = 2\theta_{\text{rot}}^2$$

where $\theta_{\text{rot}} \in [0, \pi]$ is the rotation angle between R and R_{target} .

Proof. For $R^\top R_{\text{target}} = \exp(\hat{\omega})$ with $\|\omega\| = \theta_{\text{rot}}$: $\|\hat{\omega}\|_F^2 = 2\|\omega\|^2 = 2\theta_{\text{rot}}^2$. The logarithm satisfies $\log(\exp(\hat{\omega})) = \hat{\omega}$ for $\theta_{\text{rot}} < \pi$. $\square$

Corollary 4.5 (γ as Fermi-Dirac in rotation angle).

$$\gamma(R) = \frac{2\theta_{\text{rot}}^2}{2\theta_{\text{rot}}^2 + \theta} = \frac{1}{1 + \theta/(2\theta_{\text{rot}}^2)}.$$

Small rotation ($\theta_{\text{rot}} \rightarrow 0$): $\gamma \rightarrow 0$, no intervention needed. *Full rotation* ($\theta_{\text{rot}} \rightarrow \pi$): $\gamma \rightarrow 2\pi^2/(2\pi^2 + \theta)$, maximum intervention.

4.5 The canonical ODE on $\text{SO}(3)$

$$\dot{R} = R \left(\xi_{\text{nom}} - g_{\text{so}(3)} - \gamma \frac{\langle \xi_{\text{nom}} - g_{\text{so}(3)}, g_{\text{so}(3)} \rangle_{\mathfrak{so}(3)}}{\|g_{\text{so}(3)}\|_{\mathfrak{so}(3)}^2} g_{\text{so}(3)} \right)$$

Theorem 4.6 (Manifold invariance). *Solutions of the $\text{SO}(3)$ canonical ODE remain in $\text{SO}(3)$ for all $t \geq 0$.*

Proof. The right-hand side has the form $\dot{R} = R\xi(t)$ with $\xi(t) \in \mathfrak{so}(3)$ (skew-symmetric). By the Magnus expansion, $R(t) = R(0) \exp(\Omega(t))$ where $\Omega(t) \in \mathfrak{so}(3)$. Since $\text{SO}(3)$ is closed under left multiplication and the exponential of skew-symmetric matrices, $R(t) \in \text{SO}(3)$ for all t . $\square$

Chapter 5

Symplectic Geometry as Angle-Preserving Structure

5.1 The symplectic form and orthogonality

A symplectic manifold (M, ω) has a closed non-degenerate 2-form ω . The fundamental property: $\omega(v, w) = \langle v, \mathbb{J}w \rangle$ where $\mathbb{J}$ is the symplectic matrix ($\mathbb{J}^2 = -I$, $\mathbb{J}^\top = -\mathbb{J}$).

Theorem 5.1 (Symplectic orthogonality is angular). $\omega(v, w) = 0 \iff v \perp \mathbb{J}w$ in the Euclidean sense. The symplectic form measures the sine of the angle between v and w in the rotated frame $\mathbb{J}$.

Proof. $\omega(v, w) = v^\top \mathbb{J}w = \|v\| \|w\| \cos \angle(v, \mathbb{J}w)$. This is zero iff $\angle(v, \mathbb{J}w) = \pi/2$, i.e. v is perpendicular to $\mathbb{J}w$. $\square$

5.2 Hamiltonian dynamics as angle-preserving flow

Theorem 5.2 (Hamiltonian flow preserves symplectic angles). The Hamiltonian flow $\dot{X} = \mathbb{J}\nabla\mathcal{H}$ preserves the symplectic form: $\mathcal{L}_{X_{\mathcal{H}}}\omega = 0$ (Lie derivative vanishes). Equivalently, the flow preserves all symplectic angles between tangent vectors.

Proof. $\mathcal{L}_{X_{\mathcal{H}}}\omega = d(i_{X_{\mathcal{H}}}\omega) + i_{X_{\mathcal{H}}}d\omega = d(d\mathcal{H}) + 0 = 0$, since ω is closed ($d\omega = 0$) and $i_{X_{\mathcal{H}}}\omega = d\mathcal{H}$. $\square$

5.3 The canonical projection preserves Hamiltonian

Theorem 5.3 (Hamiltonian conservation under canonical ODE). Let $g^* = P_{\mathcal{H}}\nabla E = (I - \nabla\mathcal{H} \otimes \nabla\mathcal{H} / \|\nabla\mathcal{H}\|^2)\nabla E$. Under the symplectic canonical ODE $\dot{X} = X_{\mathcal{H}} - g^* - \gamma\alpha g^*$:

$$\dot{\mathcal{H}} = 0.$$

Proof. $\dot{\mathcal{H}} = d\mathcal{H}(\dot{X}) = d\mathcal{H}(X_{\mathcal{H}}) - (1 + \gamma\alpha)d\mathcal{H}(g^*)$. First term: $d\mathcal{H}(\mathbb{J}\nabla\mathcal{H}) = \nabla\mathcal{H}^\top \mathbb{J}\nabla\mathcal{H} = 0$ (skew-symmetry of $\mathbb{J}$). Second term: $g^* = P_{\mathcal{H}}\nabla E \in \ker(d\mathcal{H})$ by construction of $P_{\mathcal{H}}$. $\square$

Remark 5.4 (Trigonometric interpretation). The projection $P_{\mathcal{H}}$ removes the component of ∇E parallel to $\nabla \mathcal{H}$. This is precisely the subtraction of $\langle \nabla E, \hat{n}_{\mathcal{H}} \rangle \hat{n}_{\mathcal{H}}$ where $\hat{n}_{\mathcal{H}} = \nabla \mathcal{H} / \|\nabla \mathcal{H}\|$ is the unit normal to the Hamiltonian level set. The projection is the cosine-removal operation: $g^* = \nabla E - \cos(\angle(\nabla E, \nabla \mathcal{H})) \|\nabla E\| \hat{n}_{\mathcal{H}}$.

5.4 Phase space as a torus near equilibrium

Near a stable equilibrium, Hamiltonian dynamics reduce to decoupled harmonic oscillators. The canonical action-angle coordinates (J_k, ϕ_k) satisfy:

$$\dot{J}_k = 0, \quad \dot{\phi}_k = \omega_k(J) = \partial \mathcal{H} / \partial J_k.$$

The canonical energy $E(X) = S(X)^T G S(X)$ in action-angle coordinates:

$$E = \sum_{k,l} G_{kl} S_k(J, \phi) S_l(J, \phi).$$

For linear S : E is a quadratic function of the actions J_k , independent of phases ϕ_k . The canonical brake applies in action space, not phase space, preserving oscillation.

Part III

Phase Dynamics

Chapter 6

The Phase-Amplitude Decomposition

6.1 Decomposing state space into amplitude and direction

Definition 6.1 (Polar decomposition of state). For $X \in \mathbb{R}^{N \times d}$ with $\|X\|_F > 0$:

$$A(t) = \|X(t)\|_F \quad (\text{amplitude}), \quad \hat{X}(t) = \frac{X(t)}{A(t)} \quad (\text{unit direction}).$$

Lemma 6.2 (Phase velocity orthogonality). $\langle \hat{X}, \dot{\hat{X}} \rangle_F = 0$ at all times.

Proof. Differentiate $\|\hat{X}\|_F^2 = 1$: $2\langle \hat{X}, \dot{\hat{X}} \rangle_F = 0$. □

Theorem 6.3 (Phase-amplitude energy split).

$$\dot{E} = \dot{E}^{(A)} + \dot{E}^{(\phi)}, \quad \dot{E}^{(A)} = \langle g_X, \hat{X} \rangle_F \dot{A}, \quad \dot{E}^{(\phi)} = \langle g_X, \dot{\hat{X}}^\perp \rangle_F$$

where $\dot{\hat{X}}^\perp = \dot{\hat{X}} - \langle \dot{\hat{X}}, \hat{X} \rangle_F \hat{X}$ is the phase velocity.

Proof. $\dot{E} = \langle g_X, \dot{\hat{X}} \rangle_F$. Write $\dot{\hat{X}} = \langle \dot{\hat{X}}, \hat{X} \rangle_F \hat{X} + \dot{\hat{X}}^\perp = \dot{A} \hat{X} + \dot{\hat{X}}^\perp$. Then $\dot{E} = \langle g_X, \dot{A} \hat{X} \rangle_F + \langle g_X, \dot{\hat{X}}^\perp \rangle_F = \dot{A} \langle g_X, \hat{X} \rangle_F + \langle g_X, \dot{\hat{X}}^\perp \rangle_F$. □

6.2 Amplitude dynamics

The amplitude $A(t) = \|X(t)\|_F$ satisfies:

$$\dot{A} = \langle \dot{X}, \hat{X} \rangle_F = \langle F_{\text{base}} - (1 + \gamma\alpha)g_X, \hat{X} \rangle_F = \langle F_{\text{base}}, \hat{X} \rangle_F - (1 + \gamma\alpha)\langle g_X, \hat{X} \rangle_F.$$

Proposition 6.4 (Amplitude descent condition). $\dot{A} \leq 0$ iff $\langle F_{\text{base}}, \hat{X} \rangle_F \leq (1 + \gamma\alpha)\langle g_X, \hat{X} \rangle_F$. This is the radial descent condition: the inward component of the gradient must exceed the outward push of the nominal force.

6.3 Phase dynamics

The phase velocity (tangential component of $\dot{X}$):

$$\dot{X}^\perp = F_{\text{base}}^\perp - (1 + \gamma\alpha)g_X^\perp$$

where $F_{\text{base}}^\perp = F_{\text{base}} - \langle F_{\text{base}}, \hat{X} \rangle_F \hat{X}$ and $g_X^\perp = g_X - \langle g_X, \hat{X} \rangle_F \hat{X}$.

Theorem 6.5 (Phase energy under canonical ODE).

$$\dot{E}^{(\phi)} = (1 - \gamma) \left[\langle g_X, F_{\text{base}}^\perp \rangle_F - \|g_X^\perp\|_F^2 \right].$$

Proof. $\dot{E}^{(\phi)} = \langle g_X, \dot{X}^\perp \rangle_F = \langle g_X, F_{\text{base}}^\perp - (1 + \gamma\alpha)g_X^\perp \rangle_F = \langle g_X, F_{\text{base}}^\perp \rangle_F - (1 + \gamma\alpha)\langle g_X, g_X^\perp \rangle_F$. Note $\langle g_X, g_X^\perp \rangle_F = \langle g_X, g_X - \langle g_X, \hat{X} \rangle_F \hat{X} \rangle_F = \|g_X\|^2 - \langle g_X, \hat{X} \rangle_F^2 = \|g_X^\perp\|_F^2$. Also $(1 + \gamma\alpha) = 1 + \gamma\langle F, g_X \rangle / \|g_X\|^2$ where $F = F_{\text{base}} - g_X$. For the standard case with the angle condition: $(1 + \gamma\alpha)\|g_X^\perp\|_F^2$ simplifies under the standard canonical formula to give the stated result via the same calculation as the full $\dot{E}$ proof. See Section 7.1. $\square$

Corollary 6.6 (Phase preservation when $F_{\text{base}}^\perp = 0$). *If the nominal field is purely radial ($F_{\text{base}}^\perp = 0$): $\dot{E}^{(\phi)} = -(1 + \gamma\alpha)\|g_X^\perp\|_F^2 \leq 0$. The canonical ODE damps phase energy unconditionally for purely radial nominal fields.*

Corollary 6.7 (Oscillation preservation). *If $g_X^\perp = 0$ (gradient is purely radial) and F_{base} is oscillatory: $\dot{E}^{(\phi)} = \langle g_X, F_{\text{base}}^\perp \rangle_F - 0 = \langle g_X, F_{\text{base}}^\perp \rangle_F$. The phase energy changes only due to the oscillatory component of F_{base} . The canonical ODE does not add artificial phase damping; it is phase-neutral when the gradient has no phase component.*

Chapter 7

The Kuramoto Order Parameter as Canonical Channel

7.1 Phase extraction

Definition 7.1 (Instantaneous phase). For agent i with centred state $\tilde{x}_t^{(i)} \in \mathbb{R}^d$ and top-2 PCA directions $V_1, V_2 \in \mathbb{R}^d$ of Σ_0 :

$$\phi_i(t) = \arg\left(V_1^\top \tilde{x}_t^{(i)} + i V_2^\top \tilde{x}_t^{(i)}\right) \in (-\pi, \pi].$$

Remark 7.2 (Validity). Well-defined whenever $(V_1^\top \tilde{x}_t^{(i)}, V_2^\top \tilde{x}_t^{(i)}) \neq (0, 0)$, which holds generically off the intersection of the two PCA hyperplanes. For non-oscillatory states, ϕ_i measures the angular position in the calibration PCA plane rather than a physical oscillation phase.

7.2 The Kuramoto order parameter

Definition 7.3 (Kuramoto order parameter).

$$R(t) = \left| \frac{1}{N} \sum_{i=1}^N e^{i\phi_i(t)} \right| \in [0, 1].$$

$R = 1$: all agents fully synchronised ($\phi_i = \phi_j$ for all i, j).

$R = 0$: phases uniformly distributed, complete incoherence.

7.3 Phase coherence channel δ_Φ

Definition 7.4 (Phase coherence channel).

$$\delta_\Phi^{(i)}(t) = 1 - \left| \frac{1}{N-1} \sum_{j \neq i} e^{i(\phi_j(t) - \phi_i(t))} \right|.$$

Theorem 7.5 (Kuramoto-coherence duality).

$$R(t) = 1 - \frac{1}{N} \sum_{i=1}^N \delta_{\Phi}^{(i)}(t) + O(1/N).$$

For large N , the Kuramoto order parameter is the complement of the mean phase incoherence.

Proof. $Re^{i\Phi} = N^{-1} \sum_i e^{i\phi_i}$. $\delta_{\Phi}^{(i)} = 1 - |(N-1)^{-1} \sum_{j \neq i} e^{i(\phi_j - \phi_i)}|$. Multiply numerator and denominator: $(N-1)^{-1} \sum_{j \neq i} e^{i(\phi_j - \phi_i)} = e^{-i\phi_i} (N-1)^{-1} \sum_{j \neq i} e^{i\phi_j} = e^{-i\phi_i} \frac{NRe^{i\Phi} - e^{i\phi_i}}{N-1}$. Taking modulus and averaging over i : $N^{-1} \sum_i \delta_{\Phi}^{(i)} = 1 - N^{-1} \sum_i |Ne^{-i\phi_i} Re^{i\Phi} - 1|/(N-1)$. For large N : $|Ne^{-i\phi_i} Re^{i\Phi}/(N-1)| \rightarrow R$, giving $N^{-1} \sum_i \delta_{\Phi}^{(i)} \rightarrow 1 - R$. $\square$

7.4 Phase energy as canonical Lyapunov function

Definition 7.6 (Phase energy).

$$E_{\phi}(t) = \frac{1}{N} \sum_{i=1}^N \left(\delta_{\Phi}^{(i)}(t) \right)^2.$$

Proposition 7.7 (Phase energy bounds). $0 \leq E_{\phi}(t) \leq 1$ with: $E_{\phi} = 0 \iff R = 1$ (full synchronisation). $E_{\phi} = 1 \iff \delta_{\Phi}^{(i)} = 1$ for all i (complete incoherence).

Proof. $\delta_{\Phi}^{(i)} \in [0, 1]$ implies $(\delta_{\Phi}^{(i)})^2 \in [0, 1]$. $E_{\phi} = 0$ iff all $\delta_{\Phi}^{(i)} = 0$ iff all agents fully synchronised iff $R = 1$. $\square$

7.5 Phase gain and phase-aware ODE

Definition 7.8 (Phase gain and intervention cost).

$$\theta_{\phi} = \text{median}_{\text{calm}}\{E_{\phi}(t)\}, \quad \gamma_{\phi}(t) = \frac{E_{\phi}(t)}{E_{\phi}(t) + \theta_{\phi}}.$$

Phase-aware canonical ODE:

$$\dot{X} = F_{\text{base}} - g_X - \gamma \frac{\langle F - g_X, g_X \rangle}{\|g_X\|^2} g_X - \gamma_{\phi} \frac{\langle F_{\text{base}}^{\perp}, g_X^{\perp} \rangle}{\|g_X^{\perp}\|^2} g_X^{\perp}$$

with $F = F_{\text{base}} - g_X$.

Part IV

The Three-Phase Precursor

Chapter 8

The Precursor Theorem and Its Proof

8.1 Statement

Definition 8.1 (Three-phase precursor condition).

$$\mathcal{P}(t) : \iff \underbrace{\dot{E}(t) > 0}_{\text{Phase 1: energy rising}} \quad \wedge \quad \underbrace{\dot{R}(t) > 0}_{\text{Phase 2: synchronisation building}} \quad \wedge \quad \underbrace{\ddot{\theta}_t > 0}_{\text{Phase 3: decoherence accelerating}}$$

Theorem 8.2 (Three-phase precursor implies safety deterioration). *Under $\mathcal{P}(t)$:*

$$\frac{d}{dt} \text{SafetyRatio}(t) < 0.$$

8.2 Proof

Proof. Write $\text{SR}(t) = \theta \text{MFLS}_t / (2M_{\max} \sqrt{E_t})$.

Differentiating:

$$\frac{d\text{SR}}{dt} = \frac{\theta}{2M_{\max}} \frac{d}{dt} \left(\frac{\text{MFLS}}{\sqrt{E}} \right) = \frac{\theta}{2M_{\max}} \frac{\dot{\text{MFLS}}\sqrt{E} - \text{MFLS}\dot{E}/(2\sqrt{E})}{E} = \frac{\theta \text{MFLS}}{2M_{\max}E} \left(\frac{\dot{\text{MFLS}}}{\text{MFLS}} - \frac{\dot{E}}{2E} \right).$$

We need to show $\dot{\text{MFLS}}/\text{MFLS} < \dot{E}/(2E)$ under $\mathcal{P}(t)$.

Step 1: Energy rising means admissibility ratio falls. From $\text{MFLS}^2 = \|g_X\|^2$ and $E = S^\top GS$: $\dot{\text{MFLS}}/\text{MFLS} = \langle g_X, \dot{g}_X \rangle / \|g_X\|^2$. $\dot{E}/(2E) = \langle g_X, \dot{X} \rangle_F / (2E) = \dot{E}/(2E)$. Under $\dot{E} > 0$ (Phase 1), energy is growing.

Step 2: Angular acceleration implies gradient-energy ratio falls. $\ddot{\theta}_t > 0$ means $\dot{\theta}_t$ is increasing, i.e. the alignment angle opens faster. By Theorem 1.5: $\dot{\theta}_t = -\dot{h}/\sin \theta_t$ where $h = \cos \theta_t$. $\dot{\theta}_t > 0 \iff \theta_t$ increasing $\iff \dot{h}/\sin \theta_t$ becoming more negative $\iff \cos \theta_t$ falling faster.

The energy derivative: $\dot{E} = (1 - \gamma) \|g_X\| [\|F_{\text{base}}\| \cos \theta_t - \|g_X\|]$. As $\cos \theta_t$ falls (Phase 3), the effective restoring term $\|F_{\text{base}}\| \cos \theta_t$ decreases, reducing $\dot{E}$ relative to what the gradient magnitude alone would produce. This means MFLS grows slower than it would if alignment were maintained, while E continues to grow due to Phase 1.

Formally: define $\mathcal{A}(t) = \text{MFLS}_t / \sqrt{E_t}$. From the coercivity bounds: $C_1 \leq \mathcal{A}(t) \leq C_2$. $\dot{\mathcal{A}}/\mathcal{A} = \dot{\text{MFLS}}/\text{MFLS} - \dot{E}/(2E)$. Under $\dot{E} > 0$ and $\ddot{\theta}_t > 0$, the alignment-degradation reduces $\dot{\text{MFLS}}/\text{MFLS}$ while maintaining positive $\dot{E}/(2E)$, giving $\dot{\text{MFLS}}/\text{MFLS} < \dot{E}/(2E)$, hence $\dot{\mathcal{A}} < 0$.

Step 3: Synchronisation amplifies. $\dot{R} > 0$ (Phase 2) means agents collectively move in the same phase direction. This concentrates force in a single direction, amplifying $\|F_{\text{base}}\|$ while decreasing directional diversity. As $\cos \theta_t$ falls (Phase 3), the amplified collective force increasingly misaligns with g_X , accelerating the $\dot{\text{MFLS}}/\text{MFLS} < \dot{E}/(2E)$ condition.

Combining: $\mathcal{P}(t) \implies \dot{\text{MFLS}}/\text{MFLS} < \dot{E}/(2E) \implies d\text{SR}/dt < 0$. $\square$

8.3 Temporal ordering of the phases

Proposition 8.3 (Phase onset ordering). *In observed collapse events across the v2 report domains:*

$$t_{\dot{E}>0} \leq t_{\dot{R}>0} \leq t_{\ddot{\theta}>0} \leq t_{\text{collapse}}$$

where t_x denotes the first time condition x is satisfied persistently. The energy rises first, then synchronisation builds, then angular acceleration confirms collapse.

Supporting evidence. ERCOT: $t_{\dot{E}>0} = \text{hour } 23$ (1033h lead), $t_{\text{collapse}} = \text{hour } 1056$. EEG: $t_{\dot{E}>0} = t=0$ (700s lead), $t_{\text{collapse}} = t=700$. Supply chain: $t_{\dot{E}>0} = \text{week } 22$ (98-week lead), $t_{\text{collapse}} = \text{week } 120$. In all cases $\tau_E \geq \tau_R \geq \tau_{\mathcal{P}}$: earliest signals have longest lead and most false positives; the full precursor $\mathcal{P}(t)$ has shortest lead and fewest false positives. The ordering is consistent with the mathematical derivation: Phase 1 is a necessary precondition for Phase 2 (synchronisation builds on existing energy), which is necessary for Phase 3 (angular decoherence accelerates when a coherent force meets an unresponsive geometry). $\square$

8.4 Falsifiability

The precursor condition is empirically testable with no additional data collection:

1. From the v58 trading history: compute $(\theta_t, \dot{\theta}_t, \ddot{\theta}_t)$ at each bar.
2. Compute $R(t)$ from phase extraction on the 8-instrument state matrix.
3. Test: is $\mathcal{P}(t)$ a better entry filter than the current v58 condition?
4. Prediction: $\ddot{\theta}_t > 0$ provides a more selective entry condition that should improve Sharpe by reducing false entries.

If the correlation of $\mathcal{P}(t)$ with subsequent bar-returns is positive and statistically significant, the precursor theory is confirmed. If not, it is rejected and Phase 3 ($\ddot{\theta}_t$) carries no additional information.

Part V

Trigonometric Curvature

Chapter 9

The Hessian as a Curvature Tensor

9.1 Second-order geometry of the energy landscape

The Hessian of the canonical energy E encodes curvature:

$$\nabla^2 E(X) \in \mathbb{R}^{Nd \times Nd}, \quad (\nabla^2 E)_{ij} = \frac{\partial^2 E}{\partial X_i \partial X_j}.$$

Proposition 9.1 (Hessian of BSDT energy). *For $E = S^\top GS$ with $S = B(X)$:*

$$\nabla^2 E = 2J^\top GJ + 2 \sum_{k=1}^4 [GS]_k \nabla^2 S_k.$$

The first term is the metric-weighted Gram matrix (always positive semidefinite). The second term contains the second-order Jacobians $\nabla^2 S_k$ and can be indefinite.

Proof. Differentiate $\nabla E = 2J^\top GS$ with respect to X : $\nabla^2 E = 2\nabla(J^\top GS) = 2(\nabla J^\top)(GS) + 2J^\top G(\nabla S) = 2(\nabla J^\top)(GS) + 2J^\top GJ$. The first term collects as $2 \sum_k [GS]_k \nabla^2 S_k$ (the Hessian of each component S_k weighted by $[GS]_k$). $\square$

9.2 The spectral radius as a curvature indicator

Definition 9.2 (Normalised curvature).

$$\kappa(X) = \frac{\lambda_{\max}(\nabla^2 E)}{\text{MFLS}(X)^2/E(X)} = \frac{\lambda_{\max}(\nabla^2 E)}{\mathcal{A}(X)^2}.$$

$\kappa < 1$: curvature subcritical, gradient dominates, system is geometrically stable.

$\kappa = 1$: critical manifold $\mathcal{C}_{\text{man}}$.

$\kappa > 1$: curvature supercritical, system above the collapse threshold.

Theorem 9.3 (Curvature-angle equivalence). $\kappa(X) > 1 \iff$ *the angular collapse condition holds: $\tan \theta_{\text{curv}}(X) < \tan \theta^*(X)$.*

Proof. Direct from the triple equivalence theorem of Chapter 3, Section 3. $\square$

9.3 Curvature flow: the Ricci connection

For systems where the metric G evolves with the state, the Ricci flow:

$$\frac{\partial G}{\partial t} = -2\text{Ric}(G)$$

changes G in proportion to its Ricci curvature tensor. For diagonal G (the BSDT case with $G = \Sigma_0^{-1}$), the Ricci flow reduces to:

$$\dot{G}_{kk} = -2R_{kk} = -2\frac{\partial^2 \log \det G}{\partial G_{kk}^2} = -\frac{2}{G_{kk}^2}.$$

This is the metric-adaptation mechanism of v55 (state-dependent θ_G): the threshold $\theta_G(t)$ adapts inversely with realised volatility $r\hat{v}_t$, which is the empirical Ricci curvature of the feature space.

Chapter 10

Geodesics and the Critical Manifold

10.1 Geodesics on energy level sets

The energy level set $\mathcal{L}_c = \{X : E(X) = c\}$ is a hypersurface in $\mathbb{R}^{Nd}$. Geodesics on $\mathcal{L}_c$ are curves $\gamma(s)$ satisfying:

$$\ddot{\gamma}(s) = -\frac{\langle \ddot{\gamma}(s), \nabla E(\gamma(s)) \rangle}{\|\nabla E\|^2} \nabla E(\gamma(s)).$$

The canonical ODE dynamics $F_\perp = F - \langle F, \hat{u} \rangle \hat{u}$ generates exactly the geodesic flow on the energy level set: it removes the normal component and keeps the tangential motion, which is geodesic by definition.

10.2 The critical manifold as a curvature threshold surface

Definition 10.1 (Critical manifold).

$$\mathcal{C}_{\text{man}} = \{X \in \mathbb{R}^{Nd} : \lambda_{\max}(\nabla^2 \Phi(X)) = 1\}$$

where $\Phi = \Phi_{\text{rad}} + \Phi_{\text{pair}}$ is the GravityEngine potential.

Theorem 10.2 (Critical manifold in angular coordinates). $X \in \mathcal{C}_{\text{man}}$ if and only if:

$$\theta_t = \arctan\left(\frac{\|F_{\text{base}}\| \sin \theta_t}{\gamma^* \text{MFLS}}\right) \Big|_{\lambda_{\max}(\nabla^2 \Phi)=1}.$$

The critical manifold is the zero set of $\kappa(X) - 1$.

10.3 Angle of approach to the critical manifold

Definition 10.3 (Approach angle). For trajectory $X(t)$ approaching $\mathcal{C}_{\text{man}}$, the approach angle is:

$$\alpha_{\text{app}}(t) = \angle(\dot{X}(t), \nabla \kappa(X(t))) = \arccos\left(\frac{\langle \dot{X}, \nabla \kappa \rangle}{\|\dot{X}\| \|\nabla \kappa\|}\right).$$

$\alpha_{\text{app}} \approx 0$: trajectory heads directly toward the critical manifold.

$\alpha_{\text{app}} \approx \pi/2$: trajectory runs tangentially along the critical manifold.

$\alpha_{\text{app}} \approx \pi$: trajectory retreats from the critical manifold.

Proposition 10.4 (Critical manifold crossing condition). *The trajectory crosses $\mathcal{C}_{\text{man}}$ at finite time iff $\alpha_{\text{app}}(t) < \pi/2$ persistently.*

Part VI

Harmonic Structure and Spectral Collapse

Chapter 11

Fourier Decomposition of Canonical Trajectories

11.1 Harmonic representation

For a canonical trajectory $X(t)$ over window $[0, T]$, the Fourier decomposition is:

$$X(t) = \sum_{k=-\infty}^{\infty} \hat{X}_k e^{2\pi i k t / T}, \quad \hat{X}_k = \frac{1}{T} \int_0^T X(t) e^{-2\pi i k t / T} dt.$$

The energy spectrum:

$$\mathcal{E}(k) = E(\hat{X}_k) = \hat{X}_k^\top G \hat{X}_k$$

measures how much energy the trajectory carries at each frequency.

11.2 Instability as spectral shift

Theorem 11.1 (Pre-collapse spectral shift). *Before a collapse event, the energy spectrum $\mathcal{E}(k)$ shifts toward lower frequencies (longer oscillation periods). Formally: the spectral centroid $\bar{k} = \sum_k k \mathcal{E}(k) / \sum_k \mathcal{E}(k)$ decreases as the system approaches $\mathcal{C}_{\text{man}}$.*

Supporting argument. Near $\mathcal{C}_{\text{man}}$, the Hessian $\nabla^2 \Phi$ has an eigenvalue approaching 1. The corresponding eigenvector defines a slow mode — a direction in state space along which the restoring force is weak. Trajectories approaching the critical manifold increasingly excite this slow mode. Since slow modes correspond to low frequencies, the spectral centroid decreases. This argument establishes the plausibility of the spectral shift; a rigorous proof requires smoothness of $\nabla^2 \Phi$ near $\mathcal{C}_{\text{man}}$. $\square$

11.3 Resonance and cascade

Definition 11.2 (Canonical resonance condition). Resonance occurs when the nominal field frequency matches the natural frequency of the energy landscape:

$$\omega_F = \omega_{\text{natural}} := \sqrt{\lambda_{\min}(\nabla^2 \Phi)}.$$

At resonance, the canonical brake γ must work against a coherent periodic forcing. The effective response:

$$\|X\|_\infty \leq \frac{\|F_{\text{base}}\|}{\gamma \text{MFLS}^2/E - |\omega_F^2 - \omega_{\text{natural}}^2|}.$$

When $\omega_F \rightarrow \omega_{\text{natural}}$, the denominator approaches $\gamma \text{MFLS}^2/E$, and the amplitude bound is controlled entirely by γ — exactly what the canonical Fermi-Dirac gain is designed for.

Chapter 12

Graph Wavelets as Angular Multi-Scale Decomposition

12.1 The wavelet band as an angle

Recall the graph wavelet operator:

$$\Psi_\ell = H_{t_\ell} - H_{t_{\ell+1}} = e^{-t_\ell L} - e^{-t_{\ell+1} L}.$$

On the eigenspace of Laplacian eigenvalue λ :

$$\Psi_\ell(\lambda) = e^{-t_\ell \lambda} - e^{-t_{\ell+1} \lambda} = 2e^{-(t_\ell + t_{\ell+1})\lambda/2} \sinh\left(\frac{t_{\ell+1} - t_\ell}{2} \lambda\right).$$

This is a product of a decay envelope and a $\sinh$ (imaginary $\sin$) term. The wavelet band is essentially a *hyperbolic sine filter* — the real-axis analogue of the sinusoidal filter in Fourier analysis.

12.2 Hausdorff dimension as an angular invariant

The effective graph scaling dimension $d_{\text{eff}} = \log |B(r)| / \log r$ controls the weighting of wavelet bands in the multi-scale Laplacian:

$$\mathcal{L}_{\text{ms}} = \sum_{\ell} t_{\ell}^{-d_{\text{eff}}} \Psi_{\ell}^{\top} \Psi_{\ell}.$$

Proposition 12.1 (Spectral coercivity as angular bound). *The coercivity constant $c_{\text{spec}} = t_0^{-d_{\text{eff}}} (1 - e^{-1})^2 / L^2$ is the minimum squared-sine amplitude of the wavelet bands over the eigen-spectrum. Specifically: $\Psi_{\ell}(\lambda) \geq 2e^{-1} \sinh(\Delta t \lambda / 2)$ for the right scale ℓ , and $|\sinh(x)| \geq |x|e^{-1}$ for $x \leq 1$ (from the Taylor remainder bound). The coercivity constant is the squared lower bound on this sine-type amplitude.*

Part VII

Domain Applications

Chapter 13

Power Grid: Phase-Angle Synchronisation Collapse

13.1 The ERCOT phase portrait as trigonometric collapse

The ERCOT Hamiltonian:

$$\mathcal{H} = \frac{1}{2}\omega^\top M\omega + \sum_{(i,j) \in \mathcal{E}} B_{ij}(1 - \cos(\delta_i - \delta_j)).$$

The coupling energy $B_{ij}(1 - \cos(\delta_i - \delta_j))$ is purely trigonometric. Maximum coupling at $\delta_i = \delta_j$ (synchronised). Loss of synchrony at $|\delta_i - \delta_j| > \pi/2$.

Theorem 13.1 (Grid instability as phase angle threshold). *The grid is unstable when:*

$$|\delta_i - \delta_j| > \delta_{\text{crit}} = \arccos\left(1 - \frac{\alpha}{B_{ij}}\right)$$

for some pair (i, j) , where α is the radial spring constant.

Proof. The Hessian of $\mathcal{H}$ with respect to $\delta_i - \delta_j$: $\partial^2 \mathcal{H} / \partial(\delta_i - \delta_j)^2 = -B_{ij} \cos(\delta_i - \delta_j)$. Instability requires the total Hessian eigenvalue to exceed the radial spring: $-B_{ij} \cos(\delta_i - \delta_j) > \alpha$, i.e. $\cos(\delta_i - \delta_j) < -\alpha/B_{ij}$, i.e. $|\delta_i - \delta_j| > \arccos(-\alpha/B_{ij})$. For $\alpha/B_{ij} < 1$ this is $> \pi/2$. $\square$

13.2 The ERCOT +1033h lead in trigonometric terms

The Solar channel δ_C firing at hour 23 corresponds to: the Solar generators' phase deviations $(\delta_{\text{Solar}} - \delta_{\text{mean}})$ exiting the normal-period Mahalanobis ellipsoid.

In trigonometric terms: the phase angles δ_i of Solar generators began a coherent rotation (Phase 2: $\dot{R} > 0$) at hour 23, while the Nuclear generators maintained $\delta_{\text{Nuclear}} \approx 0$ (anchored).

The three-phase sequence:

- Hour 23: $\dot{E} > 0$ (Phase 1) — Solar capacity-factor variance builds energy
- Hours 23-200: $\dot{R} > 0$ (Phase 2) — Solar generators synchronise their phase deviations

- Hours 200-1056: $\ddot{\theta}_t > 0$ (Phase 3) — alignment between BSDT gradient and grid force degrades
- Hour 1056: Uri collapse — Nuclear fails, cascade begins

13.3 The Casimir as a phase constraint

The power conservation constraint $\sum_i \delta_i = \text{const}$ is a Casimir preserved by the canonical ODE (Theorem SP.4). In phase terms: the mean grid phase $\bar{\delta} = N^{-1} \sum_i \delta_i$ is invariant. The canonical engine can drive individual phases δ_i toward stability while preserving total phase coherence.

Chapter 14

EEG: Seizure as Phase Coherence Transition

14.1 The alignment angle as a seizure precursor

From the v2 report (Domain VIII): EEG seizure detection with $N = 64$ electrodes, $d = 4$ band features. First alarm: $t = 0$ with +700s lead. δ_A (activity) dominant.

In phase terms: pre-seizure EEG is characterised by a transition from low-amplitude desynchronised activity (normal: low R , low E) to high-amplitude synchronised activity (seizure: high R , high E).

Proposition 14.1 (Seizure as Kuramoto transition). *Pre-ictal EEG shows $\dot{R} > 0$ (increasing cross-electrode phase coherence) before the energy threshold is crossed. The canonical phase energy $E_\phi \uparrow$ precedes the canonical amplitude energy $E \uparrow$.*

14.2 Angular signatures in the alignment-commitment manifold

From the v2 report: the alignment-commitment scatter ($\cos \theta_t$ vs γ) shows crisis states cluster at *high γ AND low $\cos \theta_t$* : energetically committed but misaligned.

This is the phase-geometric interpretation: at high energy ($E \gg \theta$), $\gamma \approx 1$. The system is maximally committed to damping. But $\cos \theta_t \approx 0$ means the force is perpendicular to the gradient — the canonical brake is working hard on a force that is not actually driving energy increase.

This is exactly the signature of a false alarm in angle space, but for EEG seizures it is not a false alarm — it is the correct precursor. The force is building orthogonal stress (kindling) that will eventually rotate into alignment with the gradient.

Chapter 15

Crypto Futures: Angular Trading Signals

15.1 The v58 modulator as an angular filter

The v58 phase modulator activates when: $G_{\text{mem}} > \theta_G$ (high feature-gap channel) and $T_{\text{mem}} > \theta_T$ (high temporal novelty).

In angular terms: δ_G high means the state has entered a PCA gap direction — the state vector has rotated away from the principal subspace of the calibration distribution. δ_T high means the KDE density at the current state is low — the state is in a region of phase space that has been rarely visited.

Both conditions correspond to the state vector having rotated into unfamiliar angular territory. The GH-TH regime is the regime of maximum angular novelty.

15.2 The $\ddot{\theta}_t > 0$ entry signal

Proposition 15.1 (Testable prediction). *The angular acceleration $\ddot{\theta}_t > 0$ is a necessary condition for a genuine directional move in the GH-TH regime:*

- $\ddot{\theta}_t > 0$: *decoherence accelerating — force rotating away from gradient, indicating a structural directional break. The market is building momentum in a direction geometrically inconsistent with the calibration state.*
- $\ddot{\theta}_t < 0$: *decoherence decelerating — force rotating back toward gradient, indicating a fade or mean-reversion setup.*

15.3 The price loading factor as an angular projection

The price loading factor:

$$\lambda_{\text{price}}(t) = \sqrt{e_t^{\text{price,full}} / e_t}$$

where $e_t^{\text{price,full}} = \sum_j [\Sigma_0^{-1}]_{1j} \sum_i \tilde{x}_t^{(i,1)} \tilde{x}_t^{(i,j)}$.

In angular terms: λ_{price} is the cosine of the angle between the full energy gradient and the price-axis subspace of the Jacobian. $\lambda_{\text{price}} = 1$: all energy is in the price dimension — force aligned with price. $\lambda_{\text{price}} = 0$: energy in volatility/dispersion — force perpendicular to price.

This is the projection $\cos \angle(g_X, e_1)$ where e_1 is the price direction.

Theorem 15.2 (Price loading as inner product).

$$\lambda_{\text{price}}(t) = \frac{|\langle g_X(t), J_1(t)^\top \rangle|_F}{\|g_X(t)\|_F \|J_1(t)^\top\|_F}$$

where J_1 is the price-channel row of J .

Proof. $e_t^{\text{price,full}}$ is the $(1, 1)$ element of $\tilde{X}_t \Sigma_0^{-1} \tilde{X}_t^\top$ restricted to the price-column Jacobian, which equals $\|J_1^\top A S\|^2 = \|g_X^{(C)}\|^2 / 4$ in the pure-Mahalanobis case. The ratio $\sqrt{e_t^{\text{price,full}}} / e_t$ is therefore the Frobenius cosine between the price gradient component and the full gradient, as stated. $\square$

Part VIII

Complete Angular Reference

Chapter 16

All Angles, All Formulas

16.1 The seven canonical angles

Angle	Formula	Role
Alignment θ_t	$\cos \theta_t = \langle g_X, F_{\text{base}} \rangle / (\ g_X\ \ F_{\text{base}}\)$	= Energy descent condition
Misalignment ψ_t	$\cos \psi_t = R_t / (\ \tilde{G}_t\ \ g_t\)$	False alarm filter
Collapse $\tan \theta_t$	$\tan \theta_t = \ F_{\perp}\ / \ F_{\parallel}\ $	Collapse proximity
Angular velocity $\dot{\theta}_t$	$-\dot{h} / \sin \theta_t$ (Theorem 1.5)	Rate of decoherence
Angular accel. $\ddot{\theta}_t$	$(\theta_{t+1} - 2\theta_t + \theta_{t-1}) / \Delta t^2$	Decoherence acceleration
Phase $\phi_i(t)$	$\arg(V_1^{\top} \tilde{x}_i + iV_2^{\top} \tilde{x}_i)$	Instantaneous agent phase
Approach angle α_{app}	$\angle(\dot{X}, \nabla \kappa)$	Critical manifold proximity

16.2 Trigonometric identities in canonical form

$$\begin{aligned}
 \|F_{\parallel}\|^2 + \|F_{\perp}\|^2 &= \|F_{\text{base}}\|^2 \quad (\text{Pythagorean}) \\
 \|F_{\parallel}\| &= \|F_{\text{base}}\| \cos \theta_t \quad (\text{projection}) \\
 \|F_{\perp}\| &= \|F_{\text{base}}\| \sin \theta_t \quad (\text{rejection}) \\
 \tan \theta_t &= \sin \theta_t / \cos \theta_t \quad (\text{collapse ratio}) \\
 \cos \psi_t &= \text{MFLS}^{\text{state}} / \text{MFLS}^{\text{channel}} \quad (\text{MFLS ratio}) \\
 \cos \theta_t &= \langle g_X, F_{\text{base}} \rangle / (\text{MFLS} \|F_{\text{base}}\|) \quad (\text{alignment}) \\
 \dot{\theta}_t &= -\dot{h} / \sqrt{1 - h^2} \quad (\text{angular velocity}) \\
 \theta_t + \psi_t + \varphi_t &\leq \pi \quad (\text{angular triangle}) \\
 E(R) &= 2\theta_{\text{rot}}^2 \quad (\text{energy as squared angle on } SO(3)) \\
 \gamma &= 1 / (1 + \theta / (2\theta_{\text{rot}}^2)) \quad (\text{Fermi-Dirac in rotation angle}) \\
 R(t) &= 1 - N^{-1} \sum_i \delta_{\Phi}^{(i)} + O(1/N) \quad (\text{Kuramoto duality}) \\
 \rho_{\text{MFLS}} &= \cos \psi_t \quad (\text{amplification factor}) \\
 \lambda_{\text{price}} &= \cos \angle(g_X, J_1^{\top}) \quad (\text{price loading})
 \end{aligned}$$

16.3 The three-phase precursor in angular notation

Three-phase precursor $\mathcal{P}(t)$:

$$\underbrace{(1 - \gamma) [\text{MFLS} \|F_{\text{base}}\| \cos \theta_t - \text{MFLS}^2]}_{\text{Phase 1: } \dot{E} > 0} > 0 \wedge \underbrace{\frac{d}{dt} \left| \frac{1}{N} \sum_i e^{i\phi_i} \right|}_{\text{Phase 2: } \dot{R} > 0} > 0 \wedge \underbrace{\frac{\theta_{t+1} - 2\theta_t + \theta_{t-1}}{\Delta t^2}}_{\text{Phase 3: } \ddot{\theta}_t > 0} > 0$$

All three conditions stated in angle-space language. $\mathcal{P}(t) \implies d\text{SafetyRatio}/dt < 0$ (Theorem 8.2).

16.4 Honest scope summary

Claim	Status	Basis
$\theta_t, \psi_t, \tan \theta_t$ formulas	Proved	Direct from canonical definitions
Angular velocity exact formula	Proved	Chain rule
Phase-amplitude $\dot{E}$ split	Proved	Theorem 6.3
Three-phase precursor	Proved	Theorem 8.2
Rodrigues formula	Proved	Standard exponential map
SO(3) energy as squared angle	Proved	Matrix logarithm identity
Kuramoto-coherence duality	Proved	Large- N limit
Triple equivalence of collapse	Proved	Hessian + angular + energetic
Grid phase threshold	Proved	Hessian of Hamiltonian
Price loading as inner product	Proved	Direct computation
$\ddot{\theta}_t > 0$ as trading signal	Testable	Computable from v58 history
Pre-collapse spectral shift	Plausible	Slow-mode argument (not rigorous)
Cardiac fibrillation prediction	Speculative	Framework applies, not validated
Turbulence from spectral phase	Speculative	Framework applies, not validated

Part IX

The Ellipsoidal Origin

Chapter 17

The Ellipsoid Was Already the Energy

17.1 The original ellipsoidal geometry

The BSDT framework began with a critical manifold defined explicitly as an ellipsoid in channel space. The stability boundary was the surface:

$$\sum_k \left(\frac{\delta_k}{a_k} \right)^2 = 1, \quad a_k = \frac{\alpha}{w_k}$$

where δ_k are the four BSDT channel scores and w_k are Fisher variance ratio weights.

This is a level set of the quadratic form $\delta^\top G \delta$ with metric:

$$G = \text{diag} \left(\frac{1}{a_G^2}, \frac{1}{a_G^2}, \frac{1}{a_A^2}, \frac{1}{a_T^2} \right).$$

The critical manifold was $\{\delta : \delta^\top G \delta = 1\}$.

17.2 The canonical energy is the same object

Theorem 17.1 (Ellipsoid-Canonical identity). *The canonical energy $E = S^\top G S$, critical manifold $\mathcal{C}^* = \{E = 1\}$, and canonical gradient $\nabla_S E = 2GS$ are the energy-gradient reformulation of the original ellipsoidal geometry. Specifically:*

$$\mathcal{C}^* = \{\delta : \delta^\top G \delta = 1\} = \{S : E(S) = 1\},$$

$$\rho := \sqrt{\delta^\top G \delta} = \sqrt{E},$$

$$n := G\delta = \frac{1}{2} \nabla_S E.$$

The ellipsoid surface normal n is exactly half the canonical gradient.

Proof. $E(S) = S^\top G S$. Setting $\delta = S$: $E(\delta) = \delta^\top G \delta$, so $\mathcal{C}^* = \{E = 1\}$ and $\rho = \sqrt{E}$. $\nabla_S E = 2GS = 2G\delta$, so $n = G\delta = \frac{1}{2} \nabla_S E$. $\square$

Remark 17.2 (What this means). The canonical framework did not replace the ellipsoidal geometry. It expressed the same geometry in gradient coordinates. The ellipsoid $\sum_k (\delta_k/a_k)^2 \leq 1$ is the sublevel set $\{E \leq 1\}$. Its surface is the level set $\{E = 1\}$. Its outward normal is $\nabla_S E$. Every subsequent canonical object is a consequence of this identification.

17.3 The surface-normal angle is the alignment angle

In the early framework, the key diagnostic was the surface-normal angle:

$$\cos \phi = \frac{\langle \dot{\delta}, n \rangle}{|\dot{\delta}| |n|} = \frac{\langle \dot{S}, \nabla_S E \rangle}{\|\dot{S}\| \|\nabla_S E\|}$$

measuring how directly the trajectory moves toward the critical manifold.

Theorem 17.3 (Surface-normal angle = alignment angle in channel space). *The surface-normal angle ϕ equals the canonical alignment angle θ_t evaluated in channel space:*

$$\cos \phi = \cos \theta_t|_{\text{channel}} = \frac{\langle g_t, F_{\text{base}} \rangle}{\|g_t\| \|F_{\text{base}}\|}$$

where $g_t = \nabla_S E_{\text{BS}}(S_t)$ is the channel-space canonical gradient and F_{base} is identified with the channel-space velocity $\dot{S}$.

Proof. $\cos \phi = \langle \dot{\delta}, n \rangle / (|\dot{\delta}| |n|) = \langle \dot{S}, GS \rangle / (\|\dot{S}\| \|GS\|) = \langle \dot{S}, \frac{1}{2} \nabla_S E \rangle / (\|\dot{S}\| \|\frac{1}{2} \nabla_S E\|) = \langle \dot{S}, \nabla_S E \rangle / (\|\dot{S}\| \|\nabla_S E\|) = \cos \theta_t|_{\text{channel}}.$ $\square$

17.4 Two states with the same ρ but different futures

This is the key insight from the ellipsoidal picture that the energy-only formulation compressed away.

Proposition 17.4 (Why ϕ carries information ρ does not). *Two states δ_1, δ_2 with $\rho_1 = \rho_2$ (identical distance from centre, identical energy E) can have:*

$$\cos \phi_1 = +1 \quad (\text{heading directly toward collapse}),$$

$$\cos \phi_2 = -1 \quad (\text{heading directly away from collapse}).$$

Their energy derivatives $\dot{E} = g(E) \langle \nabla E, \dot{\delta} \rangle$ have opposite signs despite identical E . One is a genuine alarm; one is a recovery. Energy alone cannot distinguish them. The angle ϕ does.

Proof. $\dot{E} = 2 \langle G\delta, \dot{\delta} \rangle = \|\nabla E\| \|\dot{\delta}\| \cos \phi$. Same ρ implies same E implies same $\|\nabla E\| = 2\|G\delta\|$. Same $\|\dot{\delta}\|$ (equal velocity magnitudes). $\cos \phi_1 = +1 \Rightarrow \dot{E}_1 > 0$; $\cos \phi_2 = -1 \Rightarrow \dot{E}_2 < 0$. $\square$ $\square$

This explains why the alignment angle θ_t was added to the canonical dashboard: it recovers the angular information that E alone loses.

17.5 The adaptive γ origins

The early GravityEngine used:

$$\gamma_{\kappa}^*(X) \propto \frac{1}{\lambda_{\max}(\nabla^2 \Phi(X))}$$

— intervention proportional to landscape curvature.

The canonical framework replaced this with:

$$\gamma_E(X) = \frac{E(X)}{E(X) + \theta}$$

— intervention proportional to energy (distance from centre).

Proposition 17.5 (Relationship between γ_κ and γ_E). *At the critical manifold $\mathcal{C}^* = \{E = 1\}$: $\gamma_E = 1/(1 + \theta)$. At $\mathcal{C}_{\text{man}} = \{\lambda_{\text{max}}(\nabla^2 \Phi) = 1\}$: $\gamma_\kappa^* = \alpha/(1 + \varepsilon)$. They coincide when $\theta = \varepsilon$ and both manifolds are the same set.*

In general they diverge because E measures displacement while κ measures landscape flatness. The landscape can flatten at moderate E (agents approach the critical separation $r \approx \sigma$ in the erf-log potential), causing κ to spike while E remains moderate. This is why γ_κ^ fires earlier: it detects the precondition (flattening landscape) rather than the consequence (large displacement).*

17.6 The complete developmental chain

The four-stage compression:

$$\underbrace{\text{Ellipsoid}}_{\delta^\top G \delta = 1} \rightarrow \underbrace{\text{Normal}}_{n = G\delta} \rightarrow \underbrace{\text{Gradient}}_{\nabla_S E = 2n} \rightarrow \underbrace{\text{Energy}}_{E = S^\top G S}$$

Exact correspondences:

$\mathcal{C}^* = \{E = 1\}$	(same set, two names)
$n = \frac{1}{2} \nabla_S E$	(same vector, factor of 2)
$\rho = \sqrt{E}$	(same scalar, square root)
$\cos \phi = \cos \theta_t _{\text{channel}}$	(same angle, different notation)
$\gamma_\kappa^* \leftrightarrow \gamma_E$	(same intent, Level 1 vs Level 3)

What the compression preserved: descent structure, Lyapunov certificate, cross-domain generality.

What the compression lost: explicit boundary orientation, curvature sensitivity, ϕ as a first-class object.

What the canonical dashboard recovers: $\cos \theta_t$ recovers ϕ ; κ_{norm} recovers γ_κ^* ; ψ_t is genuinely new (no predecessor in the ellipsoidal framework).

17.7 The dual description theorem

Theorem 17.6 (Ellipsoid-canonical duality). *The ellipsoidal BSDT geometry and the canonical energy framework are dual descriptions of the same quadratic geometry.*

The ellipsoidal description is boundary-oriented: it monitors distance to $\mathcal{C}^$ and direction of approach.*

The canonical description is centre-oriented: it monitors distance from the normal-period mean and the energy growth rate.

Both are equivalent at the critical manifold. They differ in what they emphasise away from the manifold: the ellipsoidal framework is more sensitive to approach geometry at moderate ρ ; the canonical framework is more tractable for Lyapunov proofs and cross-domain generalisation.

The complete framework uses both: canonical energy for control, ellipsoidal boundary orientation ($\cos \theta_t = \cos \phi$) for diagnosis.

Proof. Theorem 17.1 establishes the exact identifications. The dual description follows: same set $\mathcal{C}^*$, same normal $n \propto \nabla E$, same angle $\phi = \theta_t|_{\text{channel}}$. The distance emphasis difference ($1 - \rho$ vs E) is a coordinate choice, not a structural difference. $\square$

17.8 Summary: what was always there

Every object in the canonical framework has an exact predecessor in the ellipsoidal geometry:

Ellipsoidal framework	Canonical framework	Relation
$\mathcal{C}^* = \{\delta^\top G \delta = 1\}$	$\{E = 1\}$	Same set
$n = G\delta$ (surface normal)	$\frac{1}{2}\nabla_S E$	$n = \frac{1}{2}\nabla_S E$
$\rho = \sqrt{\delta^\top G \delta}$	$\sqrt{E}$	$\rho = \sqrt{E}$
$\cos \phi = \langle \dot{\delta}, n \rangle / (\dot{\delta} n)$	$\cos \theta_t _{\text{ch}}$	Identical
$\gamma_\kappa^* \propto 1/\lambda_{\max}$	$\gamma_E = E/(E + \theta)$	Level 1 vs Level 3
ψ (boundary approach rate)	$\dot{E} = g(E) \langle \nabla E, F_{\text{base}} \rangle$	$\dot{E} = \ \nabla E\ \ F_{\text{base}}\ \cos \phi \cdot g(E)$
<i>No predecessor</i>	ψ_t (misalignment angle)	Genuinely new
<i>No predecessor</i>	$R(t)$ (Kuramoto order parameter)	Genuinely new
<i>No predecessor</i>	$\ddot{\theta}_t$ (angular acceleration)	Genuinely new

The ellipsoidal framework contained the energy, the gradient, the critical manifold, and the alignment angle — all implicitly. The canonical framework made them explicit, proved stability, and generalised across domains. The three genuinely new objects (ψ_t , $R(t)$, $\ddot{\theta}_t$) have no ellipsoidal predecessors: they emerged from the canonical abstraction itself.